

Introduction

The beta distribution is the flexible family of probabilities on the unit interval $[0, 1]$. It is the natural distribution for proportions, success probabilities, and measurements bounded between two fixed limits. In Bayesian inference, it is the conjugate prior for the binomial and Bernoulli likelihoods.

Prerequisites

Continuous distributions, Bayesian inference basics.

Theory

A beta random variable with shape parameters $a > 0$ and $b > 0$ has PDF

$$f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}, \quad 0 \leq x \leq 1,$$

where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function.

Moments:

- $E[X] = a/(a+b)$.
- $\text{Var}(X) = ab/[(a+b)^2(a+b+1)]$.

Special cases:

- $a = b = 1$: Uniform(0, 1).
- $a = b = 0.5$: Jeffreys prior for the binomial probability.
- $a, b > 1$: unimodal density with mode $(a-1)/(a+b-2)$.
- $a < 1$ and $b < 1$: bimodal (poles at 0 and 1).

Conjugacy: given a Bernoulli/Binomial likelihood with success probability θ and a Beta(a, b) prior on θ , the posterior after observing k successes in n trials is Beta($a+k, b+n-k$).

Assumptions

The variable must be bounded between 0 and 1. For probabilities near 0 or 1, the distribution can have a pole; use care when simulating.

R Implementation

```
curve(dbeta(x, 2, 5), from = 0, to = 1, col = "steelblue", lwd = 2,
      main = "Beta densities", ylab = "f(x)")
curve(dbeta(x, 5, 2), add = TRUE, col = "#F4A261", lwd = 2)
curve(dbeta(x, 0.5, 0.5), add = TRUE, col = "#6A4C93", lwd = 2)
curve(dbeta(x, 2, 2), add = TRUE, col = "#2A9D8F", lwd = 2)
legend("top", legend = c("Beta(2,5)", "Beta(5,2)", "Beta(0.5,0.5)", "Beta(2,2)"),
      col = c("steelblue", "#F4A261", "#6A4C93", "#2A9D8F"), lwd = 2)

# Moments
a <- 2; b <- 5
c(mean = a / (a + b), var = a * b / ((a + b)^2 * (a + b + 1)))

X <- rbeta(1e5, a, b)
c(emp_mean = mean(X), emp_var = var(X))

# Bayesian update example
```

```
prior_a <- 2; prior_b <- 2
k <- 15; n <- 20
post_a <- prior_a + k; post_b <- prior_b + n - k
c(posterior_mean = post_a / (post_a + post_b),
  posterior_95_CI_lower = qbeta(0.025, post_a, post_b),
  posterior_95_CI_upper = qbeta(0.975, post_a, post_b))
```

Output & Results

mean	var
0.2857	0.0255

emp_mean	emp_var
0.286	0.025

posterior_mean	posterior_95_CI_lower	posterior_95_CI_upper
0.708	0.504	0.871

Empirical moments match theoretical. The Bayesian update with a Beta(2,2) prior and 15 successes in 20 trials gives a posterior mean of 0.71 with a 95% credible interval of (0.50, 0.87).

Interpretation

Beta distributions appear everywhere in Bayesian analysis of proportions. Reporting “posterior probability of success 0.71 (95% credible interval 0.50 to 0.87)” is a beta-based statement.

Practical Tips

- For weakly informative priors, Beta(1, 1) (uniform) or Beta(0.5, 0.5) (Jeffreys) are common defaults.
- Very large a and b correspond to strongly informative priors; check the prior predictive distribution to verify plausibility.
- Beta regression (**betareg**) models proportions in $(0, 1)$ as a function of covariates.
- Values exactly at 0 or 1 are problematic; transform via $(x(n - 1) + 0.5)/n$ or use zero-one-inflated beta.
- For a quick-look sanity check, a beta with $a = b$ is symmetric and centred at 0.5, scaling by $(a + b)$ as a concentration parameter.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes’ theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots